ORGANICS

General Chemistry I, Lecture Series 11

Qixi Mi

Reading:

OGB8 §7

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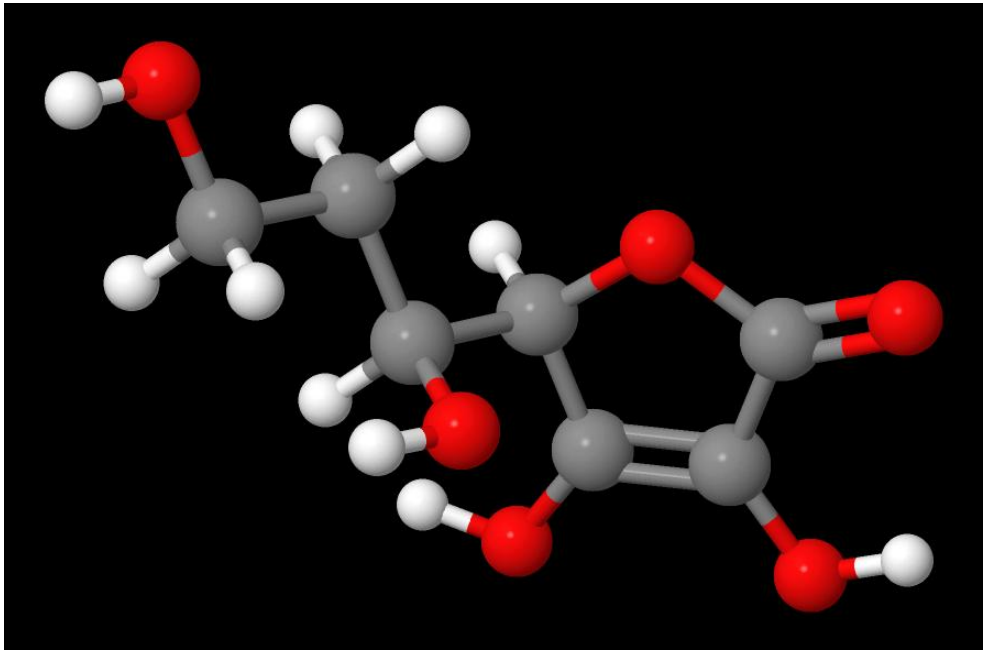
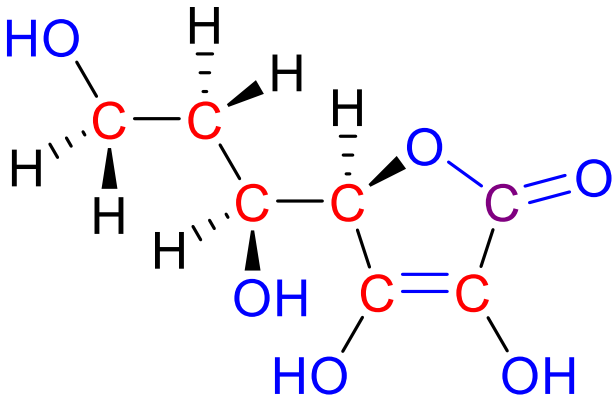
Outline

- The carbon chain
- Functional groups 官能团
- Interaction between functional groups





Ascorbic Acid 抗坏血酸
(Vitamin C 维生素C)

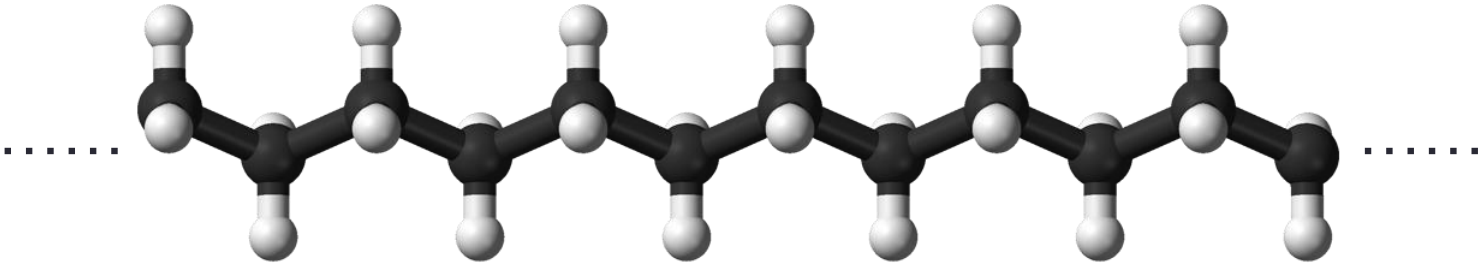
Why Carbon?

Organic Chemistry:
The study of the compounds of carbon.

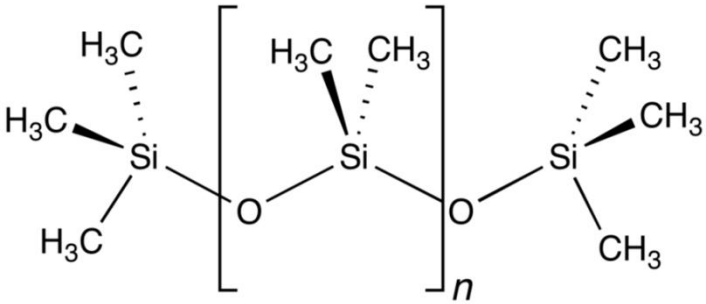
Average Bond Dissociation Energies ($kJ \cdot mol^{-1}$)

H—H	C—H	C—C	N—N	O—O	F—F	Si—O	P—O
436	414	347	161	146	153	460	377

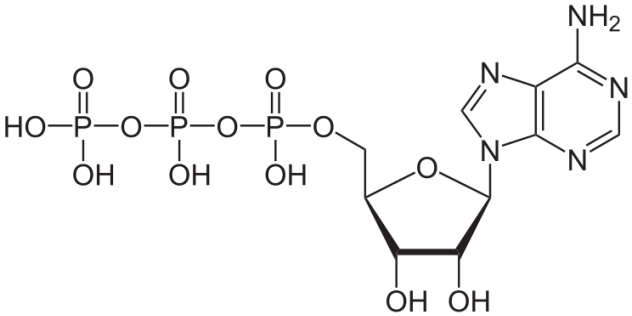
Long, Long Chains



Polyethylene 聚乙烯

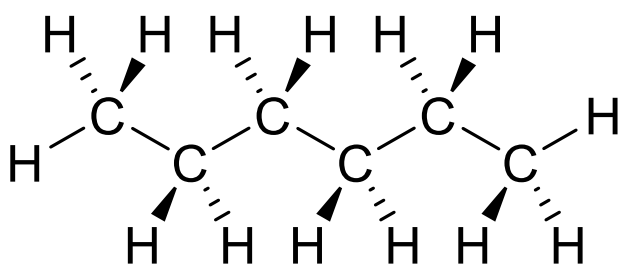


Silicone 硅酮

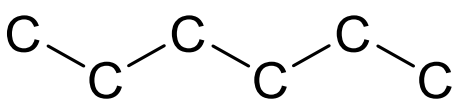


Adenosine triphosphate
三磷酸腺苷

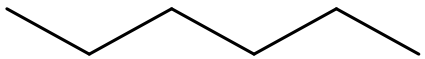
Alkanes 烷烃



$CH_3CH_2CH_2CH_2CH_2CH_3$



$CH_3(CH_2)_4CH_3$, C_6H_{14}



OGC7 p.309

n	Name
1	Methane 甲烷
2	Ethane 乙烷
3	Propane 丙烷
4	Butane 丁烷
5	Pentane 戊烷
6	Hexane 己烷
7	Heptane 庚烷
8	Octane 辛烷
9	Nonane 壬烷
10	Decane 癸烷
11	Undecane 十一烷
12	Dodecane 十二烷
13	Tridecane 十三烷
14	Tetradecane 十四烷
15	Pentadecane 十五烷

Etymology

烃 tīng = 碳 + 氢

hydrocarbon

烷 alkane C_nH_{2n+2}

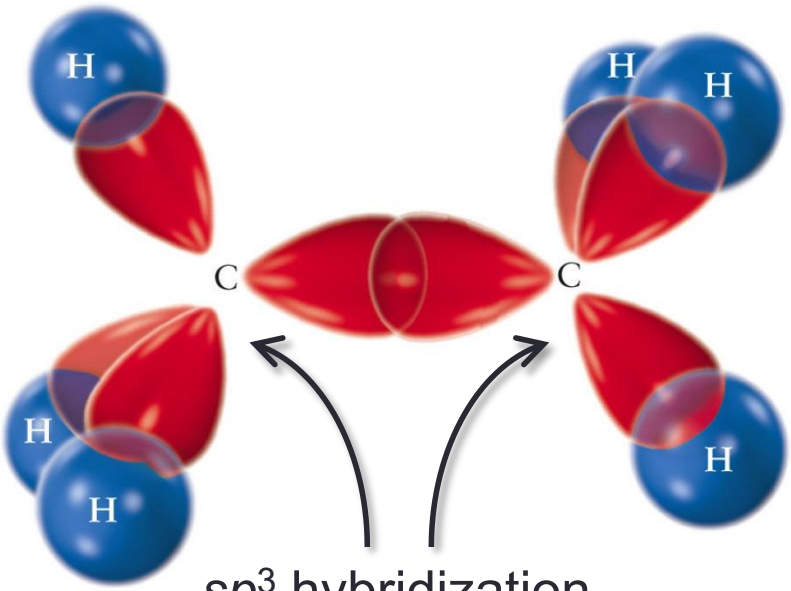
烯 alkene C_nH_{2n}

炔 alkyne C_nH_{2n-2}

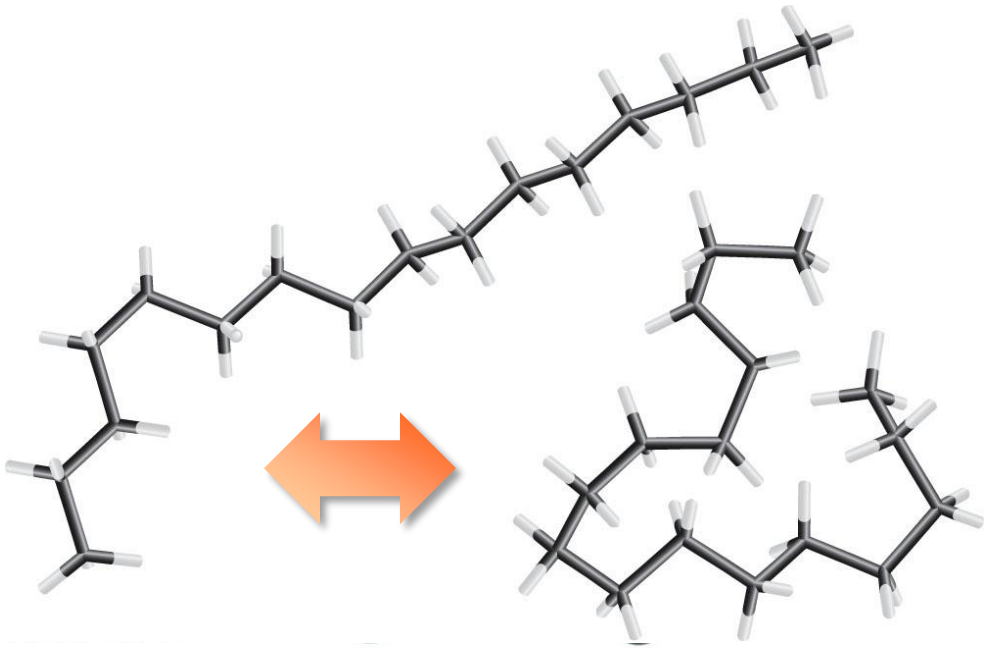
明朝那些事儿

一、太祖派
1、周系 临安王 朱勤烷（7）
2、楚系 楚庄王 朱孟烷（2）
3、蜀系 永川王 朱悦烯（2）
4、岷系 安昌王 朱定烷（7）
5、唐系 唐靖王 朱琼烃（2）
6、伊系 伊简王 朱颙炔（2）
二、成祖派
赵系 赵王 朱慈炔（12）
三、仁宗派
郑系 郑恭王 朱厚烷（7）
四、宪宗派
衡系 玉田王 朱厚烃（7）

Alkane (2)



sp^3 hybridization



Different conformations 构象

Butane C_4
B.p.沸点 ~0°C

Diesel C_{10}–C_{22}
B.p. 180–370°C

Vaseline C_{20}–C_{30}
M.p. ~37°C

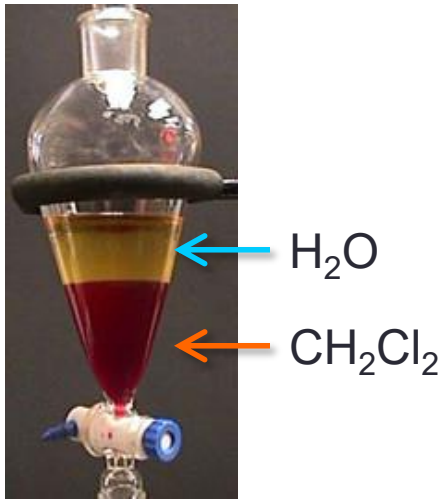
HDPE $C_{\sim 100000}$
M.p. ~130°C

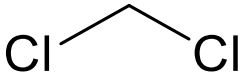
Haloalkanes 卤代烷烃

Halogen 卤素

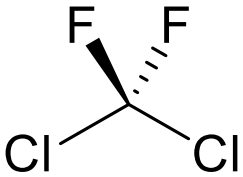
Average Bond Dissociation Energies ($kJ \cdot mol^{-1}$)

C—H	C—F	C—Cl	C—Br	C—I
414	485	339	285	213

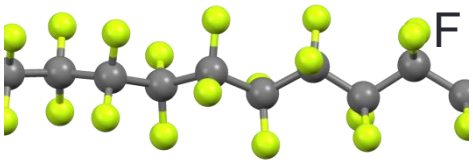




Dichloromethane
二氯甲烷



Dichlorodifluoromethane
二氟二氯甲烷
(Freon 氟利昂)

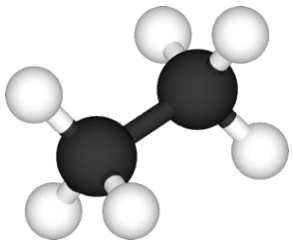


Polytetrafluoroethylene
聚四氟乙烯
(Teflon 特氟龙)

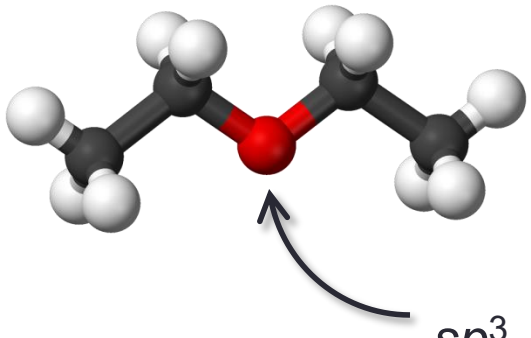
Ether 醚

Average Bond Dissociation Energies ($kJ \cdot mol^{-1}$)

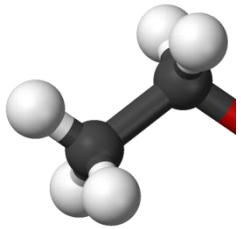
C—H	C—N	C—O
414	305	358



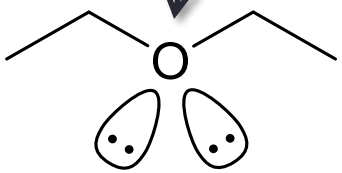
Ethane 乙烷



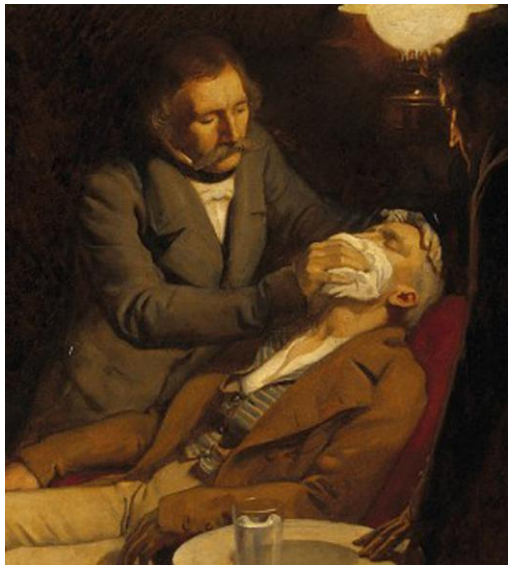
sp^3



Ethyl 乙基



(Di)ethyl ether
(二)乙(基)醚



Anesthesia 麻醉术

(Halo)alkane & Ether: Summary

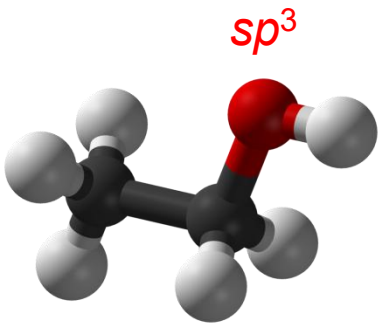
sp^3 hybridization

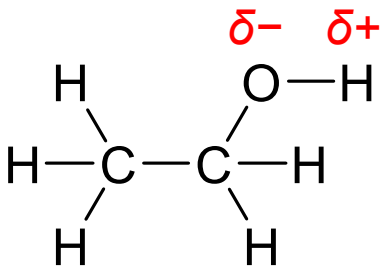
- No lone pairs or empty orbitals on C
- Stable at room temperature
- Reactive at high temperatures

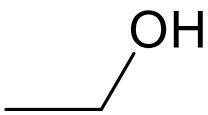
Low polarity 极性

- Lipophilic 亲油性
- Low solubility 溶解度 in H_2O
- Anesthetic

Alcohol 醇

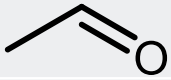
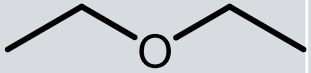
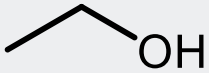






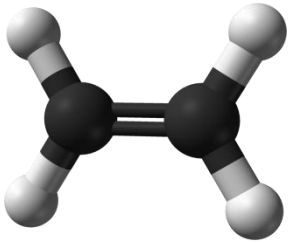
Ethanol; ethyl alcohol;
EtOH 乙醇

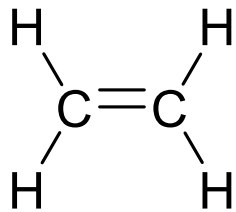
Polar solvent
极性溶剂

	B.p. (°C)
	11
	20
	35
	78

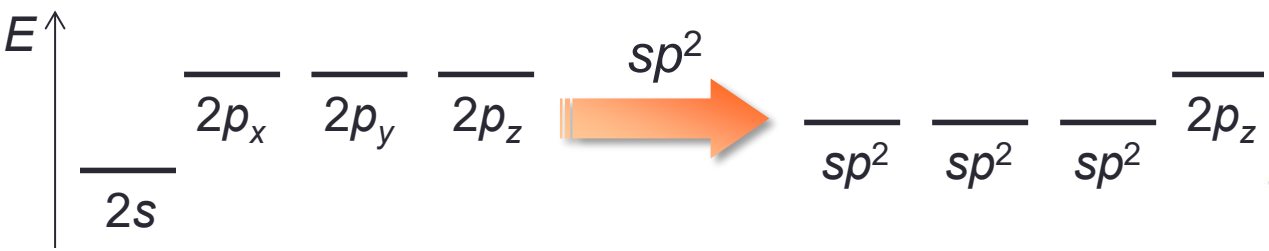
High b.p. due to
hydrogen bond

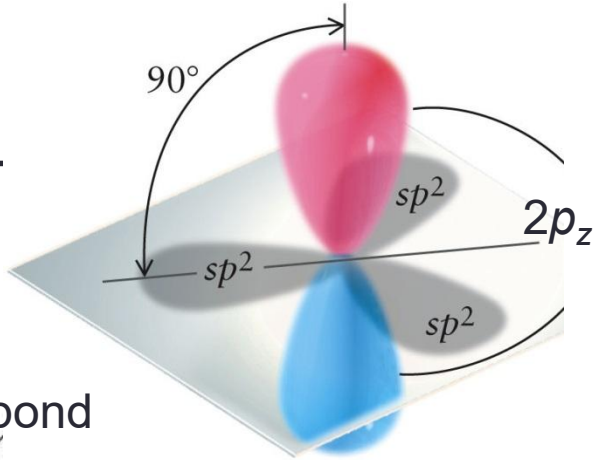
Alkene 烯烃

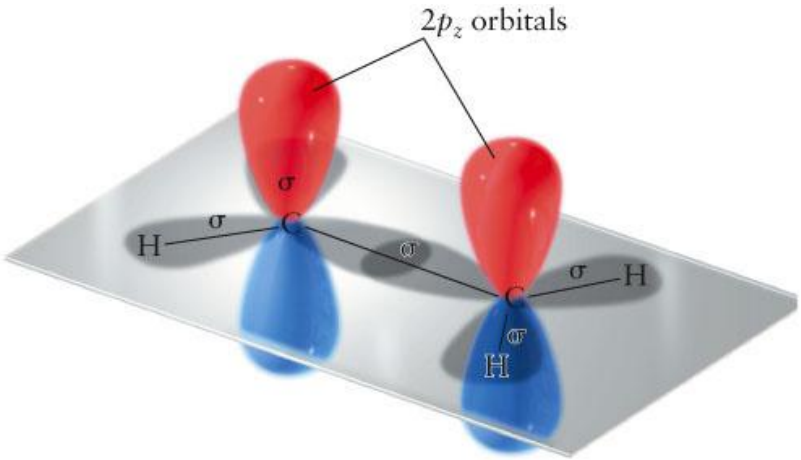


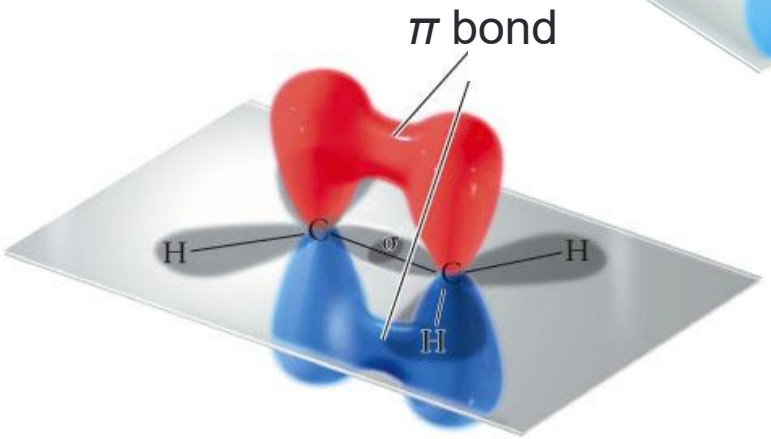


Ethene; ethylene
乙烯







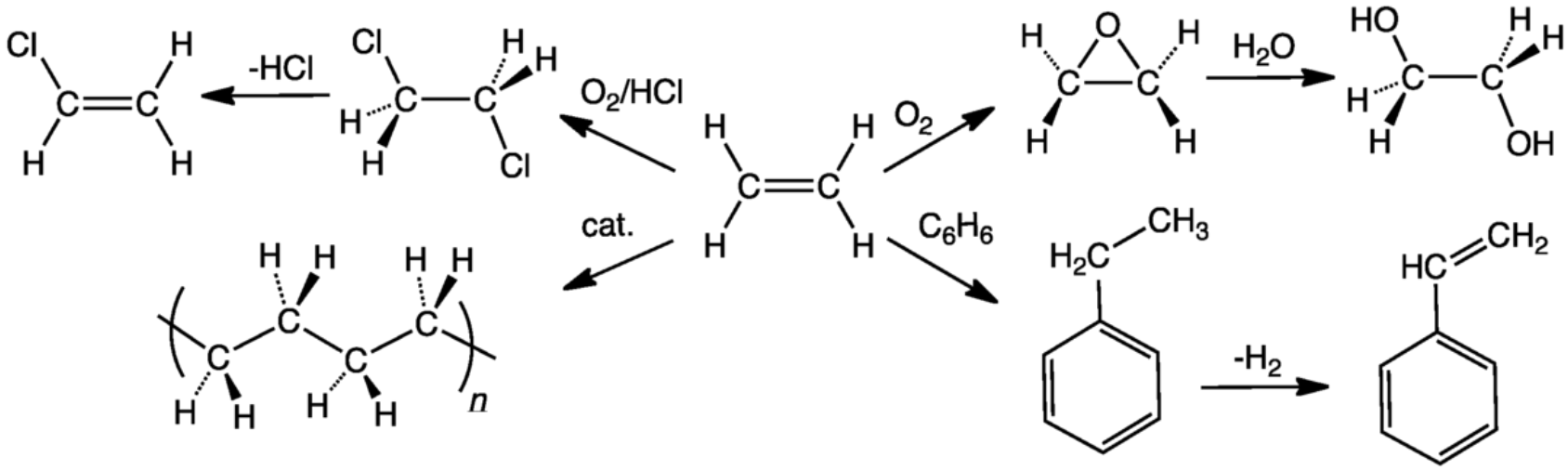


Alkene (2)

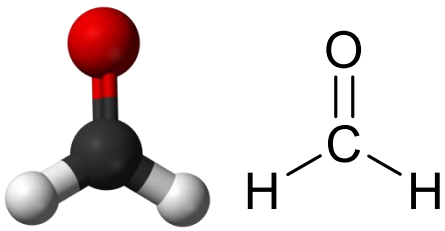
Average Bond Dissociation Energies ($kJ \cdot mol^{-1}$)

C—H	C—C	C=C
414	347	611

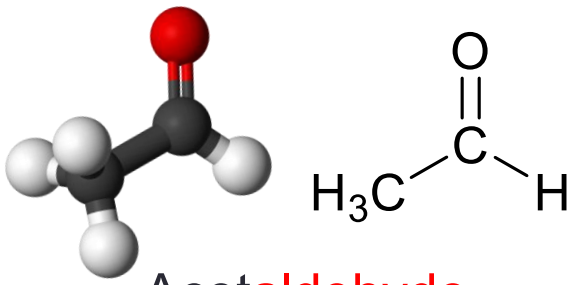
加成反应 Addition reaction:



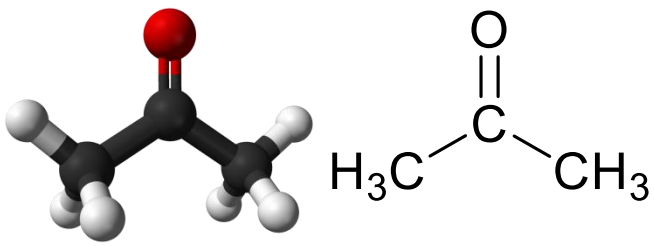
Aldehyde 醛 and Ketone 酮 (1)



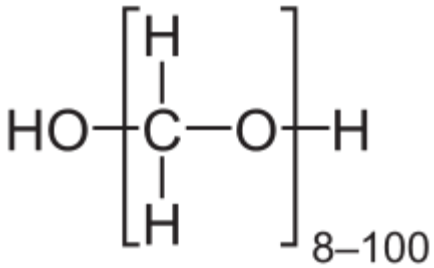
Formaldehyde
Methanal
甲醛



Acetaldehyde
Ethanal
乙醛

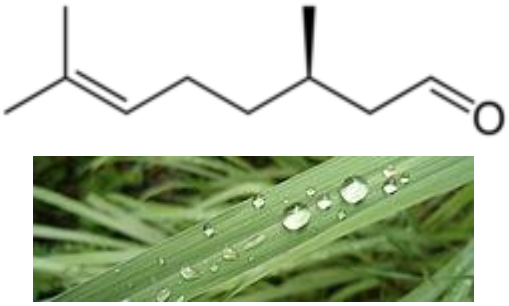


Acetone
Propanone
丙酮



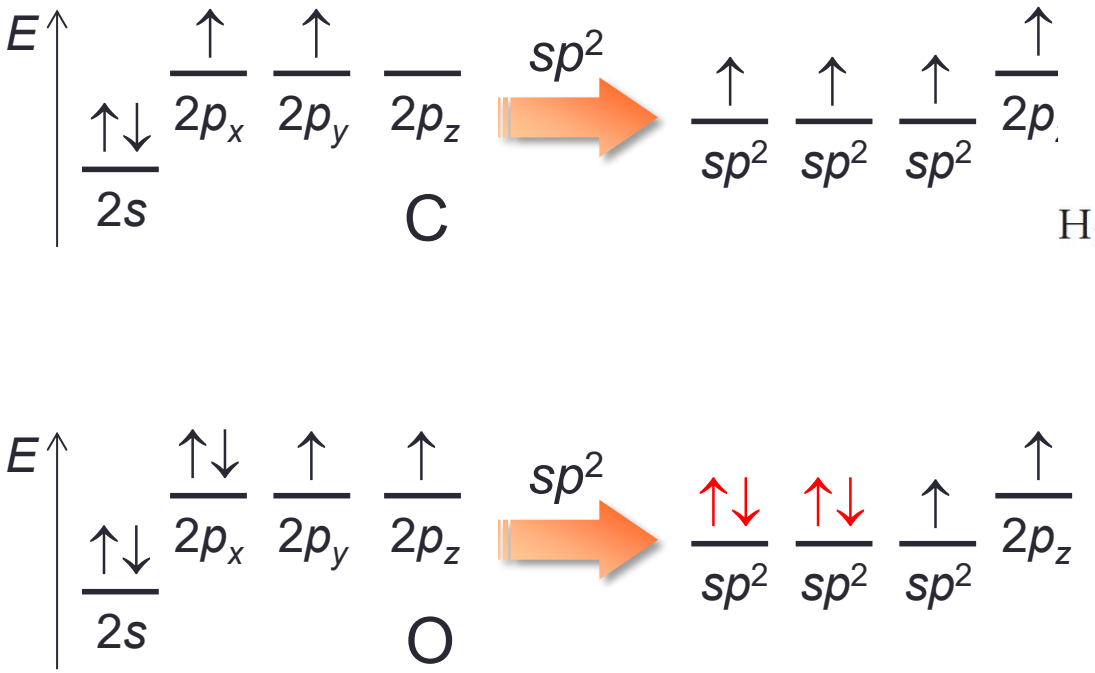
Paraformaldehyde
多聚甲醛

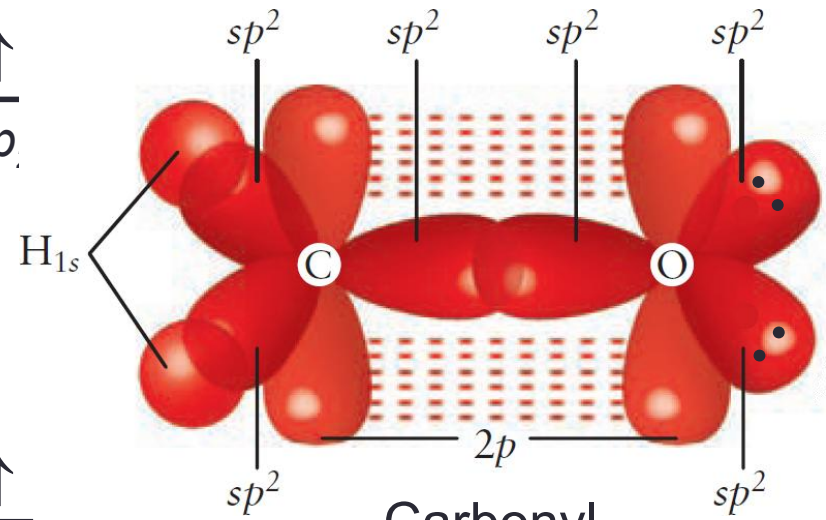
Cinnamaldehyde
肉桂醛



Citronellal
香茅醛

Aldehyde and ketone (2)

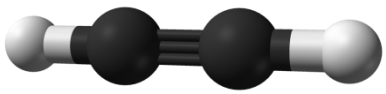




Carbonyl
羰基

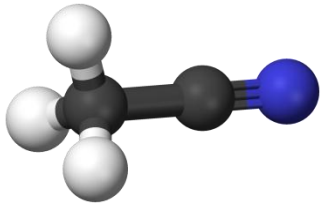
羰 tāng = 碳 + 氧

Alkyne 炔 and Nitrile 腈jīng



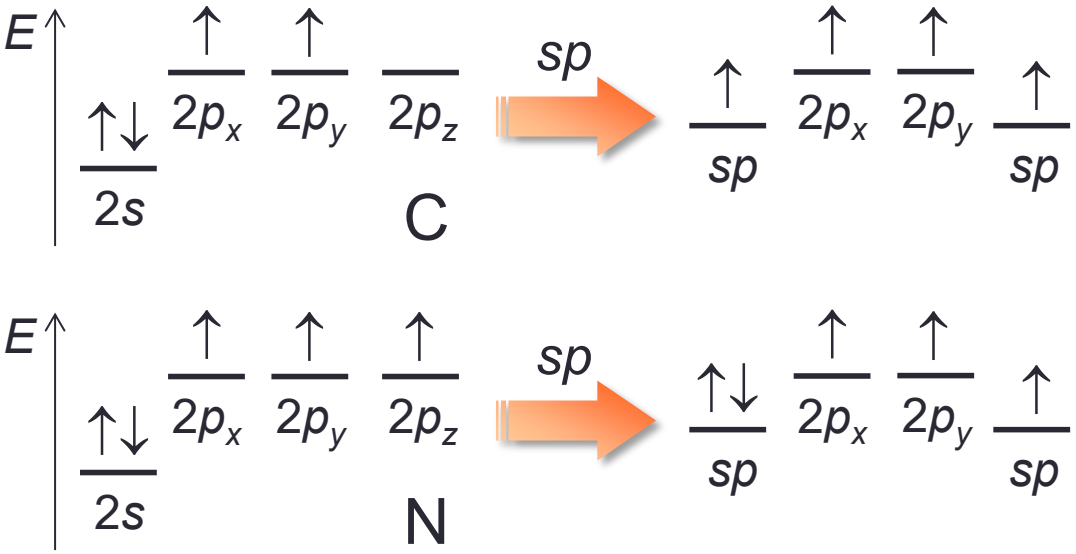
H—C≡C—H

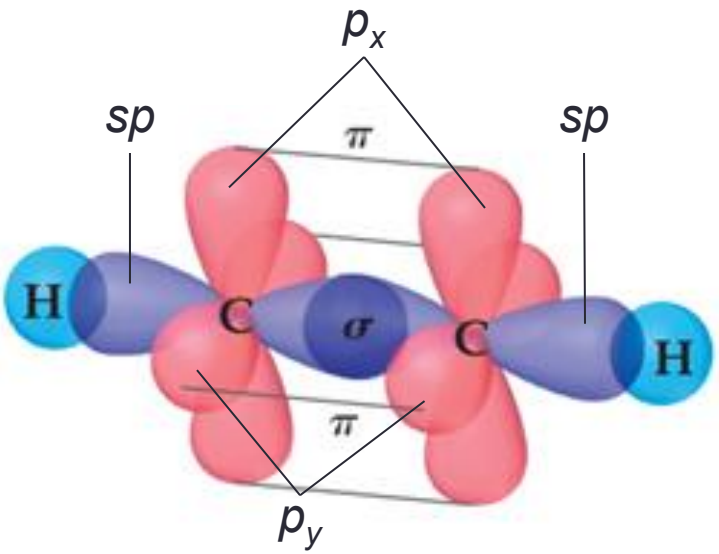
Acetylene
Ethyne
乙炔



H_3C—C≡N

Acetonitrile
Ethanenitrile
乙腈





Double and Triple Bonds: Summary

sp^2 hybridization

- 1 σ bond, 1 π bond;
  π bond has higher energy
- π bond cannot rotate
- Addition reaction
- C=C is nonpolar;
  C=O is polar.

sp hybridization

- 1 σ bond, 2 π bond;
  π bonds have higher energy
- Addition reaction
- C≡C is nonpolar;
  C≡N is polar.